

- 2 (a) (i) (the loudspeakers) are connected to the same signal generator B1 [1]
- (ii) 1. the waves (that overlap) have phase difference of zero or path difference of zero and so
 either constructive interference
 or displacement larger B1 [1]
2. the waves (that overlap) have phase difference of $(n + \frac{1}{2}) \times 360^\circ$ or $(n + \frac{1}{2}) \times 2\pi$ rad or path difference of $(n + \frac{1}{2})\lambda$ and so
 either destructive interference
 or displacements cancel/smaller B1 [1]
3. the waves (that overlap) are in phase or have phase difference of $n360^\circ$ or $2\pi n$ rad or path difference of $n\lambda$ and so
 either constructive interference
 or displacement larger B1 [1]
- (b) time period = 0.002 s or 2 ms C1
- wave drawn is half time period B1
- amplitude 1.0 cm (same as Fig. 2.2) B1 [3]

Page 3	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – October/November 2015	9702	23